



Oracle Cloud Infrastructure Data Science Professional

Empowering data scientists and ML engineers with comprehensive OCI skills for the complete machine learning lifecycle - from workspace setup to production deployment.

Comprehensive Course Overview



End-to-End ML Lifecycle

Master every phase from workspace configuration to model deployment with hands-on experience.



MLOps Best Practices

Implement industry-standard MLOps workflows for automated, scalable machine learning operations.



OCI Ecosystem Integration

Leverage Vault, Object Storage, Data Flow, and Generative AI services for comprehensive solutions.



Real-World Applications

Build business-specific ML solutions that address practical industry challenges and requirements.

OCI Data Science Environment Setup

Core Components

Workspace Management

Configure and manage workspaces, projects, and notebook sessions for collaborative development.

JupyterLab & Conda

Utilize JupyterLab environments with Conda for reproducible, isolated development workflows.

Security & Version Control

Integrate OCI Vault for secure credential management and Git repositories for code collaboration.



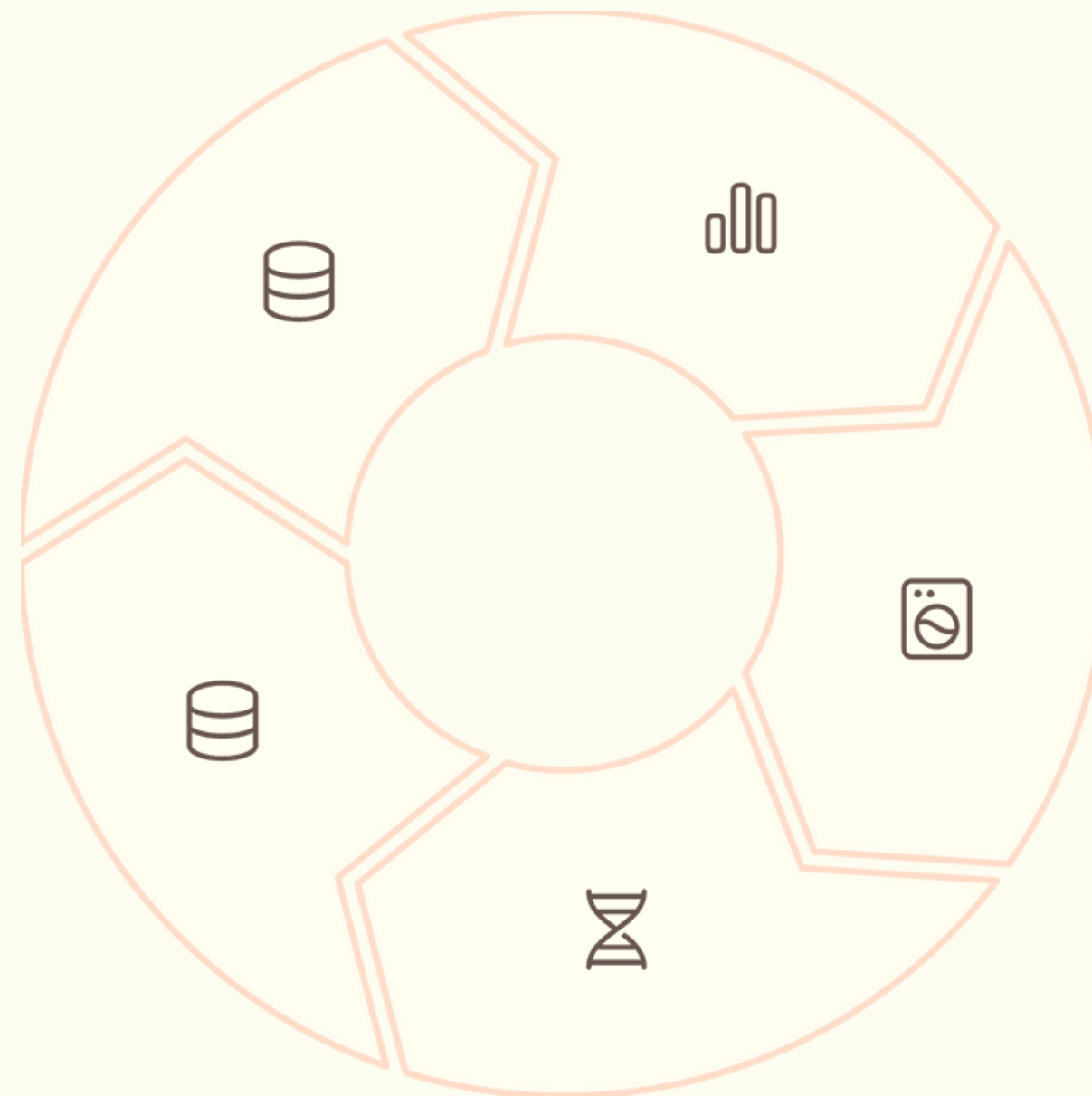
Machine Learning Lifecycle Mastery

Data Access

Connect and preprocess data from various OCI sources efficiently.

Cataloging

Organize and serialize models in the Model Catalog for reusability.



Visualization

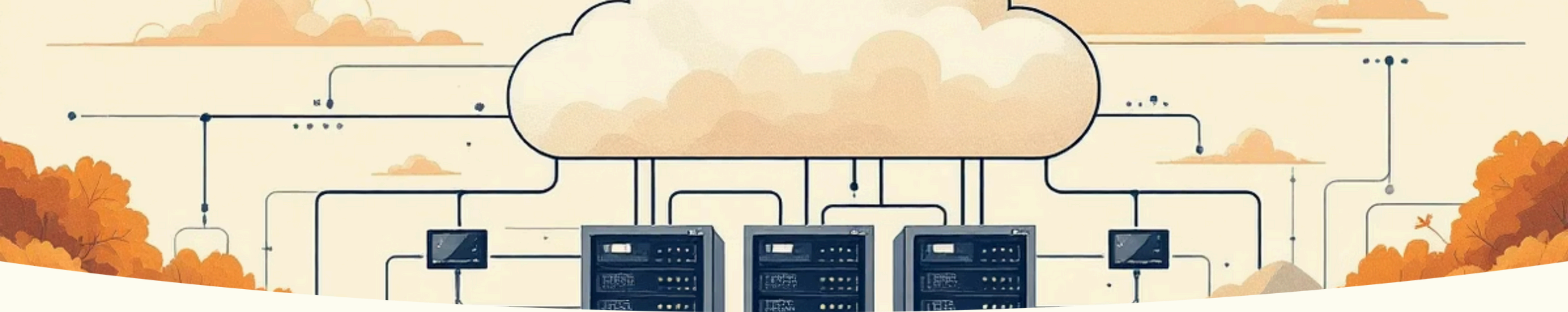
Create insightful visualizations for data exploration and analysis.

Model Training

Leverage Oracle AutoML and ADSTuner for optimized model development.

Evaluation

Assess models using global and local explainers for comprehensive insights.



Production Model Deployment

Multi-Framework Support

- TensorFlow models
- Hugging Face transformers
- LangChain RAG applications

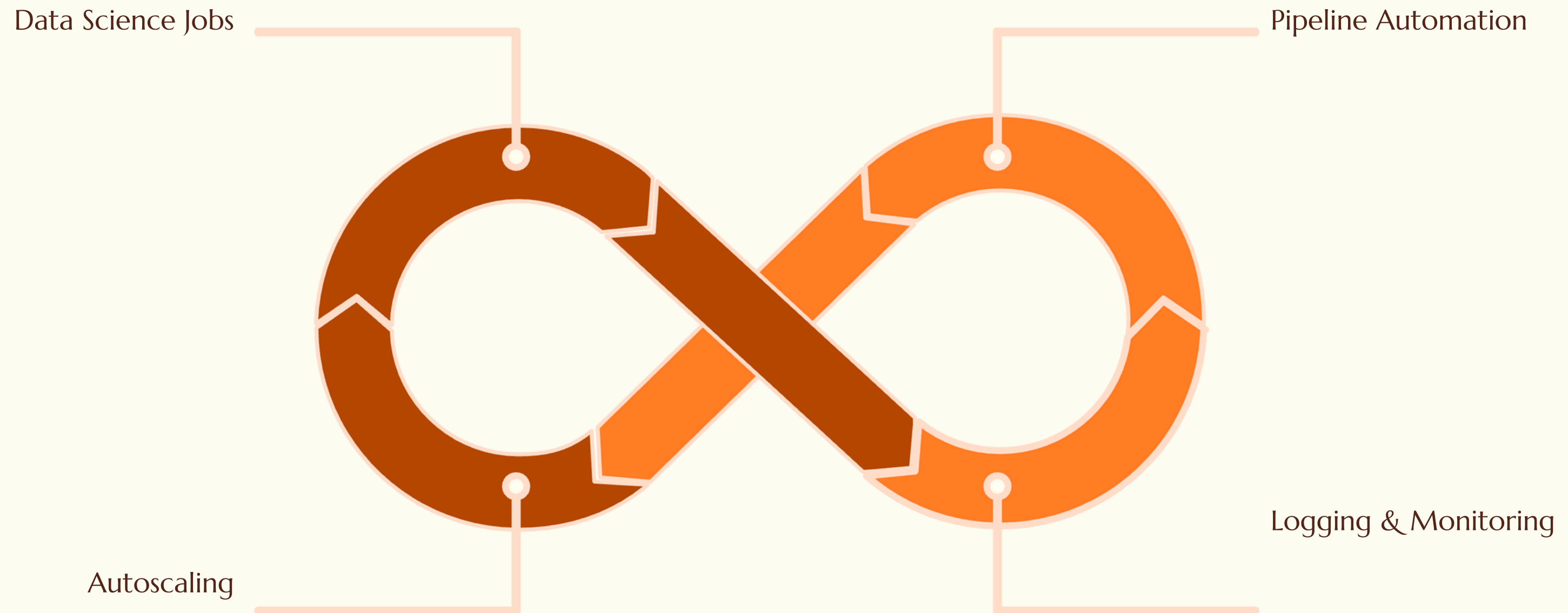
Scaling Solutions

- Automatic scaling based on demand
- Load balancing optimization
- Resource management

Quick Actions

- Streamlined deployment workflows
- Custom operators
- Rapid iteration cycles

Advanced MLOps Architecture



Implement comprehensive MLOps practices that ensure scalable, monitored, and automated machine learning operations across your organization.

Integrated OCI Services Ecosystem



OCI Data Flow

Execute large-scale Spark applications for distributed data processing and advanced analytics workflows.



Data Labeling Service

Streamline data annotation processes with built-in labeling tools and quality assurance features.



Generative AI Integration

Incorporate cutting-edge generative AI capabilities directly into your machine learning pipelines.



Open Data Access

Leverage publicly available datasets and seamlessly integrate external data sources for enriched models.

Transform Your Data Science Practice

3x

Faster
Deployments

Reduce time-to-production with automated MLOps workflows

100%

Scalable
Solutions

Enterprise-grade infrastructure that grows with your needs

99.9%

Security &
Reliability

Bank-level security with OCI's robust cloud infrastructure

Ready to revolutionize your ML workflow? Apply these OCI Data Science tools in real-world scenarios and deliver measurable business value through faster, more secure, and scalable machine learning solutions.



Thank You

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